INCH-POUND

A-A-55709
16 August 1996

COMMERCIAL ITEM DESCRIPTION

PUMP, INTRAVENOUS INFUSION

The General Services Administration has authorized the use of this Commercial Item Description for all Federal Agencies.

1. SCOPE. This Commercial Item Description covers a portable, battery operated, rechargeable, volumetric intravenous infusion pump.

2. SALIENT CHARACTERISTICS :

2.1. Shall be a portable, rechargeable, battery operated, volumetric intravenous infusion pump, suitable for the safe administration of fluid therapy.

2.2 Shall be possible to recharge the battery while the pump is in operation.

2.3 Shall accurately measure the fluid administration by the volumetric method.

2.4 Shall be capable of preselecting a volume to be administered from 1 to 999 mls. The rate of flow should at least range from 1 ml per hour to 200 hours, in 1 ml increments.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-MBP, 2800 South 20th Street, Philadelphia, PA 19145-0530, by using the self-addressed Standardization Document Improvement Proposal, (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515

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2.5 Shall have the following alarms and alerting mechanisms:

a. Audible and visible air-in-line alarm.

b. Visible low battery alarm that will operate one hour prior to need for recharging battery.

c. An occlusion alarm to indicate occluded tubing or clogged filter.

d. An audible alarm to indicate completion of infusion.

2.6 Shall be capable of attachment to the upright portion of a standard intravenous solution pole.

2.7 Standard administration sets may be used between the solution sets or as secondary sets with compatible tubing.

2.8 With the exception of the disposable cassette and plunger assembly and associated tubing, the pump should be capable of being cleaned by commonly accepted methods.

2.9 Shall be provided with features designed to prevent and/or lessen the effects of intentional removal and/or accidental removal of the set/cassette to protect against uncontrolled gravity infusions.

2.10 Varying fluid container height, the maximum distance allowed by the set tubing above and below the pump, shall not cause the flow error to be greater than (>) 5 percent.

2.11 Voltage characteristics shall permit operation on any of the following single phase, nominal voltage sources as specified by the end user:

- 115 VAC ± 10 percent, 50 Hz ± 2 Hz
- 115 VAC ± 10 percent, 60 Hz ± 2 Hz
- 230 VAC ± 10 percent, 50 Hz ± 2 Hz
- 230 VAC ± 10 percent, 60 Hz ± 2 Hz

2.11.1 Equipment which operates on multiple frequency, 50/60 Hz is acceptable. Shall have a power on/off switch and there shall be a clear, visual indication of line power connection and operation.

2.12 The pump shall be provided with a rechargeable, integral battery that shall provide, as a minimum, 4 hours of normal operation at a flow rate of 125 ml/hr.

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2.13 A visual and/or audible low-battery signal shall be activated at least 30 minutes prior to degradation of pump performance. A depleted battery shall be accompanied by an audible alarm signaling interruption of infusion. The low-battery signal shall precede the battery discharged alarm.

2.14 The battery shall have continuous recharge capability. With the unit connected to line power and operating at 125 ml/hr, it shall go from full discharge to 100 percent charge within 24 hours. With the power switch "OFF" and the unit connected to line power, it should go from full discharge to 100 percent charge within 14 hours. There shall be a clear, visual indication of line power connection to signify battery charge when the pump switch is off.

2.15 Equipment leakage current for equipment powered by a nominal 110 VAC source shall not exceed the more stringent of the limits established in the ANSI-AAMI ESI Standard for Safe Current Limits for Electromechanical Apparatus, or NFPA 99, Safety Standard for Healthcare Facilities, when measured as prescribed therein. For equipment powered by a nominal 220 VAC Source, leakage current shall not exceed those limits established in IEC-601-1, Safety of Medical Equipment.

2.16 The equipment offered, should in so far as practical, be in accordance with applicable Guidelines and Preferred Practice for the Design of Medical Equipment. Nothing in this standard shall be considered as limiting the selection of hardware, materials, or processes to the specific items described therein. Unless otherwise stated in specific provisions, this standards is applicable to the design of equipment for both men and women.

2.17 The pump shall be easy to set-up, operate and correct alarms. The unit shall provide digital displays exhibiting amount infused, rate to be infused, and cumulative infusion volume for both primary and secondary infusions.

2.18 Units shall have a tamper proof feature to prevent patient from changing the flow rate or turning off the unit. This may be accomplished by providing a locking device, programmable lock-out mode or other means of preventing unauthorized tampering.

2.19 If the device's front panel provides push button, keypad or membrane type touch sensors for data entry, adequate separation shall be provided for activation by a single finger. Activation of the data entry device should provide an audible/tactile feedback. Invalid key depression may be indicated by an audible tone.

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2.20 Labels or markings shall be durable, and shall clearly and concisely identify the function of all controls and indicators. Simple operating sequence instructions and important precautions shall be printed on or permanently displayed on the equipment.

2.21 Shall be designed to pass Aeromedical Certification requirements and must be in compliance with the following standards. Military standards shall be interpreted as though all units will be used aboard Military aircraft.

2.21.1 Electromagnetic Interference (EMI) . The purpose of the EMI tests is to verify Compliance with MIL-STD-461D and MIL-STD-462D, "Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference", and "Measurement of EMI Characteristics," respectively.

2.21.1.1 Radiated Emissions (RE-102) - This test measures radiated emissions generated by the unit. Excessive emissions could interfere with aircraft navigation and communication equipment. The unit is tested while operating on each possible power source that it is configured for, such as internal rechargeable battery, 110VAC/60Hz, 110VAC/400Hz, and 28VDC.

2.21.1.2 Radiated Susceptibility (RS-103) - This test determines whether the ambient electromagnetic fields encountered in flight interfere with the operation of the unit. During exposure to the electromagnetic induction fields the unit is operated on each possible power source that it is configured for, such as internal rechargeable battery, 110VAC/60Hz, 110VAC/400Hz, and 28VDC.

2.21.1.3 Conducted Emissions (CE-102) - This test measures emissions generated by the unit and conducted back up the power line. Excessive conducted emissions could affect the aircraft power supply and/or other systems powered from it. The unit is tested while operating on each possible power source that it is configured for, such as 110VAC/60Hz, 110VAC/400Hz, and 28VDC, except for internal battery.

2.21.1.4 Conducted Susceptibility (CS-101) - This test verifies that the unit will safely operate during allowable fluctuations in the aircraft's power supply. During the exposure the unit is tested while operating on each possible power source that it is configured for, such as 110 VAC/60 Hz, 110 VAC/400 Hz, and 28 VDC, except for internal battery.

2.21.1.5 Conducted Susceptibility (CS-114) - This test verifies that the unit will safely operate during current fluctuations along its cables. During the exposure, the unit is tested while operating on each possible power source that it is configured

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for, such as 110 VAC/60Hz, 110 VAC/400Hz, and 28 VDC, except for internal rechargeable battery.

2.21.1.6 Conducted Susceptibility (CS-115) - This test verifies that the unit will safely operate during transients in the aircraft's power supply. During the exposure, the unit is tested while operating on each possible power source that it is configured for, such as 110 VAC/60Hz, 110 VAC/400Hz, and 28 VDC, except for internal rechargeable battery.

2.21.2 Vibration . MIL-STD-810E, Environmental Test Methods and Engineering Guidelines, are followed to test the unit's construction, durability, performance, and the integrity of the mounting device while withstanding random vibration and sinusoidal vibration on the X, Y, and Z axes.

2.21.3 Environmental . MIL-STD-810E, the numerical values listed below do not come from any MIL-STD or other source document. They are modified versions of those in MIL-STDs, and simulated operational conditions. Environmental testing is performed under the following test conditions:

2.21.3.1 Humidity Operation: 94 ±4 percent relative humidity, 85°F ±3.6°F (29.5°C ±2°C) for four hours.

2.21.3.2 Hot Temperature Operation: 120 °F ±3.6°F (49°C ±2°C) for two hours.

2.21.3.3 Cold Temperature Operation: 32 °F ±7.2°F (0°C ±4°C) for two hours.

2.21.3.4 Hot Temperature Storage: 140 °F ±3.6°F (60°C ±2°C) for six hours.

2.21.3.5 Cold Temperature Storage: -40 °F ±3.6°F (-40°C ±2°C) for six hours.

2.21.4 Altitude . MIL-STD-810E, each unit is subjected to reduced atmosphere pressures from 522.7 mmHg, 10,000 ft equivalent in a hypobaric chamber. The exposure time varies with the equipment item under evaluation, but is usually a minimum of 1 hour.

2.21.5 Rapid Decompression . Decompressions in flight are uncommon; however, if one were to occur, the unit should not present a hazard to the patient, crew or aircraft operations. Each unit is subjected to a series of rapid decompression (RD) tests, starting at an equivalent pressure altitude of 8,000 ft.

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with subsequent changes of altitude to 40,000 ft over periods of 60, 7, and 1 second.

2.21.6 Airborne Feasibility . Primary objectives of this phase are to identify problems with the unit that could not be evaluated in the lab, assess human factors in the aircraft environment (in accordance with MIL-STD-1472D), assess ease of loading the unit on the aircraft, ensure visibility of visual alarms and audibility of audible alarms, check aircraft interface and unit securing methods, and the effect of the combined stresses of flight on the unit.

2.21.7 Military Standards in salient characteristic 2.21, shall be performed at:

Armstrong Laboratory/CFTS
Aeromedical Research Function
2504 D. Drive, Suite 1
Brooks AFB, Texas 78235-5104

2.22 Aeromedical Certification. The pump shall be designed to pass Aeromedical Certification requirements and must be in Compliance with the following standards. Military standards shall be interpreted as though all units will be used aboard military aircraft.

2.22.1 Electrical Safety Evaluations . The unit will not exceed the leakage current and ground resistance limits in AFI 41-203, Electrical Shock Hazards, which references National Fire Protection Agency (NFPA) 99.

2.22.2 Environmental Testing to MIL-STD-810E .

TEST METHOD	TEST CRITERIA
501.3 High Temperature Operation	49 ^{O}C/15 percent RH/2 hrs

Procedure II
The medical device will operate correctly at temperatures up to and including 49 ^{O}C.

501.3 High Temperature Storage	71 ^{O}C/15 percent RH/6 hrs

Procedure I
The medical device will operate after being stored in a high temperature environment (71 ^{O}C).

502.1 Low Temperature Operation	-25 ^{O}C/na/2 hrs

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Procedure II
The medical device will operate correctly at -25 °C.

502.1 Low Temperature Storage -46 °C/na/6 hrs

Procedure I
The medical device will operate correctly after being stored in a low temperature environment (-46 °C).

507.3 Humidity 29.5 °C/95 percent RH/4 hrs

Procedure I
The medical device will operate correctly in a high humidity and high temperature environment.

500.3 Altitude 15,000 ft. for 1 hr.

The medical device will operate correctly in a low pressure environment (15,000 foot altitude equivalent).

514.4 Vibration 2.116 Grams each Axis for 1 Hr

Procedure
The medical device will operate correctly in a vibration environment. Currently tested in each of three different axes in accordance with the procedure noted.

2.22.3 Electromagnetic Characteristics Testing to MIL-STD-461C limits:

Method	Frequency and/or V/m
CE03	30Hz - 50kHz
CE03	10kHz - 50kHz
CE04	50kHz - 50MHz

The medical device shall not produce conducted emissions in excess of current MIL-STD-461C limits.

CS02	50kHz - 400MHz
CS06	Spikes of 200 V/m

The medical device shall not be susceptible to MIL-STD-461C test generated power spikes or signals induced on its power lines.

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RE02 14kHz - 12.4GHz
The medical device shall not produce radiated emissions in excess of current MIL-STD-461C limits.

RS03 10kHz - 10GHz (20V/m)
The medical device shall not be susceptible to radiated test fields produced to MIL-STD-461C levels.

2.22.4 Electromagnetic characteristics testing is conducted in accordance with current MIL-STD-462 methods to current MIL-STD-461C limits. Testing includes applicable conducted emissions and susceptibility as well as radiated emissions and susceptibility testing.

2.22.5 In-Flight Testing . A flight evaluation can only be performed after laboratory evaluations are completed and an airworthiness release is obtained from ATCOM.

2.22.6 The medical device shall operate according to the performance test criteria in the aircraft without limitations due to environmental factors, including noise. The operator will be able to operate the medical device through it's full range of functions while wearing the flight ensemble, including gloves and a flight helmet. The medical device shall not interfere with any aircraft system. The medical device shall not be susceptible to interference from aircraft systems.

2.22.7 Military Test Standards in salient characteristic 2.22, shall be performed at:

U.S. Army Aeromedical Research Laboratory (USAARL)
SGRD-UAD-IE
Ft. Rucker, Alabama 36362.

2.23 Electrical characteristics shall be consistent with the requirements contained in the latest edition of specification MIL-D-42048.

2.24 Shall be provisioned in accordance with the requirements contained in the latest edition of specification MIL-D-42048. Specifically provisioning code “D” to include:

DI-ILSS-80865	Medical Support Kit Lists
DI-ILSS-80868	Special Equipment Tools and Test Equipment List
DPSC-A	Special Equipment Commercial Service Manual

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DPSC-B	Special Equipment Commercial Operating/ Instruction
DI-V-7000A	Supplementary Provisioning/Technical Documentation
DI-V-7009A	Design Change Notice
DI-ILSS-80867	Special Equipment, Consumable/Durable Support Item List

2.25 Workmanship. The intravenous infusion pump shall be free from defects which detract from its appearance or impair its serviceability.

2.26 Unit. Each (EA). One intravenous infusion pump, as specified, constitutes one unit. Each unit shall be packaged in accordance with the requirements of ASTM D 3951.

3. REGULATORY REQUIREMENTS.

3.1 Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

3.2 Recovered materials. The offeror/contractor is encouraged to use recovered materials in accordance with the Federal Acquisition Regulation Subpart 23.403 to the maximum extent practical.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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4.2 Metric products . Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units. Dimensions and tolerances specified shall be converted using conversion tables contained in the latest revision of ASTM E 380.

4.3 Market acceptability . The following market acceptability criteria are necessary to document the quality of the product to be provided under this Commercial Item Description.

4.3.1 The company producing the item must have been producing a product meeting the requirements of this Commercial Item Description for at least two (2) years.

4.3.2 The company must have sold 1,000 units meeting this Commercial Item Description in the commercial market place over the past two (2) years.

4.4 Other quality assurance provision . Shall be as indicated in Section E, Inspection, of the contract and/or order.

5. PACKAGING .

5.1 Preservation, packaging, packing, labeling and marking . Shall be as specified in the contract and/or order. Specific requirements for this item are indicated in Section C, Description/Specification; general packaging and marking requirements are indicated in Section D, Packaging and Marking.

6. NOTES .

6.1 National Stock Number Data . The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-01-381-4456	PUMP, INTRAVENOUS INJECTION

6.2 Source of Documents :

6.2.1 Government Documents

Defense Printing Service Detachment Office
Bldg. 4D (NPM-DODSSP)
700 Robbins Avenue
Philadelphia, PA 19111-5092

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6.2.2 Non-Government Documents .

ASTM documents.

(Applications for copies should be addressed to the American Society for Testing and Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

ANSI documents.

(Application s for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036-8002)

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS
GSA-FSS-FCGC

Project Number: 6515-5794
Location: PUMPC54.DOC (Word 6)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-55709	960816

3. DOCUMENT TITLE
PUMP, INTRAVENOUS INFUSION

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-M	(1) Commercial (215)737-8431	(2) AUTOVON 444-8431
c. ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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